$\theta_2 = 90° + \theta_1$

But in ΔABC, $\theta_2 = \angle C + \theta_1$

Therefore, $\angle C = 90°$

DO YOU KNOW?

In any triangle, exterior angle is equal to sum of the interior opposite angles.

Note

Let l_1 and l_2 be two lines with well-defined slopes m_1 and m_2 respectively, then

(i) l_1 is parallel to l_2 if and only if $m_1 = m_2$.

(ii) l_1 is perpendicular to l_2 if and only if $m_1 m_2 = -1$.

Example 5.8 (i) What is the slope of a line whose inclination is $30°$?

(ii) What is the inclination of a line whose slope is $\sqrt{3}$?

Solution (i) Here $\theta = 30°$

Slope $m = \tan\theta$

Therefore, slope $m = \tan 30° = \frac{1}{\sqrt{3}}$

(ii) Given $m = \sqrt{3}$, let θ be the inclination of the line

$\tan\theta = \sqrt{3}$

We get, $\theta = 60°$

Thinking Corner

The straight lines X axis and Y axis are perpendicular to each other. Is the condition $m_1 m_2 = -1$ true?

Example 5.9 Find the slope of a line joining the given points

(i) $(-6,1)$ and $(-3,2)$ (ii) $\left(-\frac{1}{3},\frac{1}{2}\right)$ and $\left(\frac{2}{7},\frac{3}{7}\right)$ (iii) $(14,10)$ and $(14,-6)$

Solution

(i) $(-6,1)$ and $(-3,2)$

The slope $= \frac{y_2 - y_1}{x_2 - x_1} = \frac{2-1}{-3+6} = \frac{1}{3}$.

(ii) $\left(-\frac{1}{3},\frac{1}{2}\right)$ and $\left(\frac{2}{7},\frac{3}{7}\right)$

The slope $= \frac{\frac{3}{7} - \frac{1}{2}}{\frac{2}{7} + \frac{1}{3}} = \frac{\frac{6-7}{14}}{\frac{6+7}{21}}$

$= -\frac{1}{14} \times \frac{21}{13} = -\frac{3}{26}$.

(iii) $(14,10)$ and $(14,-6)$

The slope $= \frac{-6-10}{14-14} = \frac{-16}{0}$.

The slope is undefined.

Progress Check

Fill in the missing boxes

S.No.	Points	Slope
1	$A(-a,\ b),\ B(3a,-b)$	
2	$A(2,\ 3),\ B(_,_)$	2
3	$A(_,\ _),\ B(_,_)$	0
4	$A(_,\ _),\ B(_,_)$	undefined

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